MILITARY SPECIFICATION

MICROCIRCUITS MEMORY, DIGITAL, CMOS,
4096 BIT, STATIC RANDOM ACCESS MEMORY (SRAM),
BULK SILICON AND SILICON ON SAPPHIRE

This amendment forms a part of MIL-M-38510/245, dated 28 January 1987, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

TITLE: Delete and substitute as shown above.

1.4, Input low (V_{IL}) voltage range, device types 01, 02, 03, 04:
Delete "-0.3 V".

1.4, Input high (V_{IH}) voltage range, device types 01, 02, 03, 04:
Delete "+0.3 V".

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
11	11
12	Reprinted without change
13	Reprinted without change
14	14
15	15
16	16
51	51
52	52

PAGE 2

3.2, delete and substitute:

"3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein. At present epoxy die bonding may be performed using qualified conductive silver paste or qualified (nonepoxy) silver filled glass. All epoxies must be qualified to method 5011 of MIL-STD-883". Laser scribing shall be allowed only for SOS technology product and only to the back side of the wafer.

PAGE 4

4.2e, delete second sentence and substitute: "Conditions shall be static, T_A = 125°C, supply voltage at 6.5 V minimum, and a duration of 48 hours minimum for classes S and B. The stress test circuit is figure 3, static I, all ones pattern."

4.2.f(5), delete "7 ±.5" and substitute "6 ±.5".

4.2g(4), delete "7 ±.5" volts and substitute "5.5 ±.5 volts for device types 01 through 04, 6.5 volts minimum for device types 05 and 06".

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PG 52

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AMENDMENT 1

4.2, add the following paragraph:

"i. The alternate screening procedures of MIL-STD-883, method 5004, shall be permitted as approved by the qualifying activity".

PAGE 5

4.4.2b, delete in its entirety and substitute: "ESDS testing in accordance with method 3015 of MIL-STD-883, shall be performed at initial qualification and after any process or design changes that affect the input protection circuits."

PAGE 7

TABLE I, I_{DDOP}, maximum limits column, device type 05: Delete "4" and substitute "4.5."

PAGE 8

TABLE I, t_{ELQV}, maximum limits column, device type 06: Delete "200" and substitute "280".

TABLE I, t_{ELQX}, minimum limits column, device types 01, 02: Delete "10" and substitute "5".

TABLE I, t_{ELQX}, maximum limits column, device types 03, 04: Delete "100" and substitute in minimum limits column "50".

PAGE 9

TABLE I, t_{WLQZ}, test column: Add "1/".

TABLE I, t_{WLQZ}, device type column, device type 04: Delete "1/".

PAGE 10

TABLE I, t_{DVWH}, minimum limits column, device type 06: Delete "75" and substitute "105".

TABLE I, t_{AVQZ}, symbol column, delete "1/" and add "1/" to test column.

PAGE 28

FIGURE 7, t_{ELQX}, minimum column, device types 01, 02: Delete "10" two places and substitute "5" two places.

PAGE 29

FIGURE 7, t_{ELQV}, maximum column, device type 06: Delete "200" and substitute "280".

FIGURE 7, t_{DVWH}, minimum column, device type 06: Delete "75" and substitute "105".

FIGURE 7, t_{ELQX}, maximum column, device types 03 and 04: Delete "100" two places and substitute in minimum column "50" two places.

PAGE 31

TABLE III, PWR DWN and ADD COMP, symbol column: Add "16/" two places.

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PAGE 32

TABLE III, TEST 76, minimum columns: Delete "10" three places and substitute "5" three places.

PAGE 35

TABLE III, test 94, minimum columns: Delete "10" three places and substitute "5" three places.

TABLE III, PWR DWN and ADD COMP, symbol column: Add "16/" two places.

TABLE III, test 90, measured terminal column: Delete "Q" and substitute "V_{DD}".

PAGE 38

TABLE III, PWR DWN and functional tests, symbol column: Add "16/" two places.

PAGE 41

TABLE III, PWR DWN and functional tests, symbol column: Add "16/" two places.

TABLE III, test 90, measured terminal column: Delete "Q" and substitute "V_{DD}".

PAGE 44

TABLE III, I_{DDOP}, test limits, maximum column: Delete "4.0" and substitute "4.5".

PAGE 48

TABLE III, t_{ELQV}, test limits, maximum column, T_C = +25°C and T_C = -55°C: Delete "160", two places and substitute "200", two places.

TABLE III, t_{ELQV}, test limits, maximum column, T_C = +125°C: Delete "200" and substitute "280".

PAGE 49

Footnotes 10/a and 11/a: Delete "6" and substitute "7".

Footnote 11/c, delete "7" and substitute "5".

PAGE 50

Add the following new footnote: "16/ Tested, but no values are printed for this parameter in variable data".

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1203)

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TABLE II. Electrical test requirements.

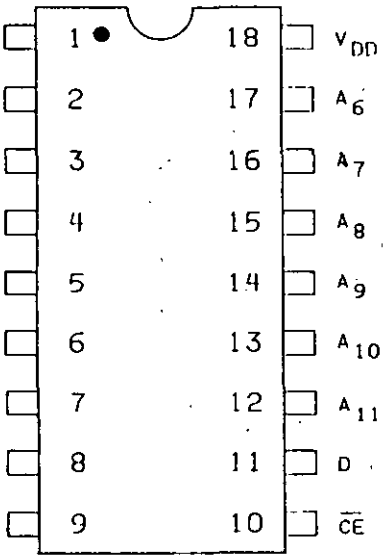
Line no.	MIL-STD-883 test requirements	Class S device 1/ 2/				Class B device 1/ 2/			
		Ref par.	Table 3/ III subgroups	Table IV delta 4/ limits	Test circuit figure	Ref par.	Table 3/ III subgroups	Table IV delta 4/ limits	Test circuit figure
1	Interim electrical parameters method 5004	4.2e	1			4.2e	1		
2	Static burn-in I method 1015	4.2b,e 4.2f(5) 4.5.2			3		Not required		
3	Same as line 1		1*	Δ					
4	Static burn-in II method 1015	4.2f(5) 4.5.2			3	4.2b,g(4) 4.5.2	Not required		
5	Same as line 1	4.2c	1*	Δ					
6	Dynamic burn-in method 1015	4.2b,f(6) 4.5.2	5/		4		5/ Required		4
7	Same as line 1	4.2c	1*	Δ		4.2c	1*	Δ	
8	Final electrical parameters method 5004		1*,2,3, 8,9,10,11				1*,2,3, 8, 9		
9	Group A test requirements method 5005	4.4.1	1,2,3,4, 8,9,10, 11,12			4.4.1	1,2,3,4, 9,10,11		
10	Group B test requirements method 5005	4.4.2 4.5.3	1,2,3,8, 9,10,11	Δ					
11	Group C end-point electrical parameters method 5005					4.4.3 4.5.3	1,2	Δ	
12	Group D end-point electrical parameters method 5005	4.4.4	1,2,3			4.4.4	1,2		
13	Group E test requirements method 1019	4.5.6	Table VI		Table VII	4.5.6	Table VI		Table VII

1/ Blank spaces indicate tests are not applicable.
2/ For subgroups 9, 10, and 11, only the worst value measured per device need be recorded when variables data is required (e.g. during qualification).
3/ (*) indicates PDA applies to subgroup 1 (see 4.2.1).
4/ (Δ) indicates delta limits shall be required, and delta values shall be computed with reference to either the initial recorded electrical parameters (see 4.5.3), or to the previous interim electrical parameters, as indicated to the qualifying activity.
5/ The device manufacturer may, at his option, either perform delta measurements or within 24 hours after burn-in (removal of temperature or bias) perform the final electrical parameter measurements, subgroup A1.

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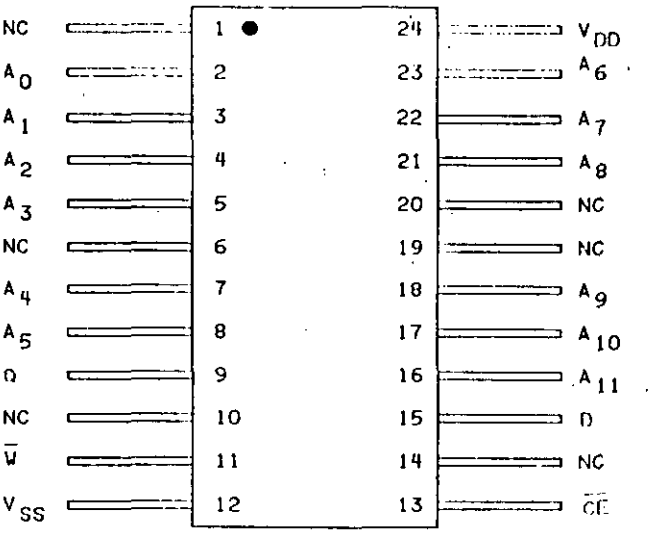
Device types 01, 03, and 05

Case V



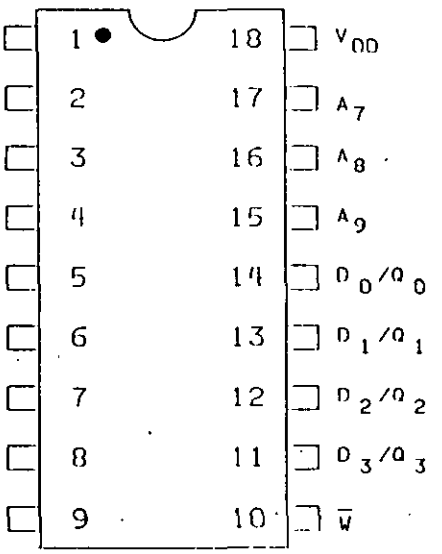
Device types 03 and 05

Case K



Device types 02, 04, and 06

Case V



Device types 04 and 06

Case K

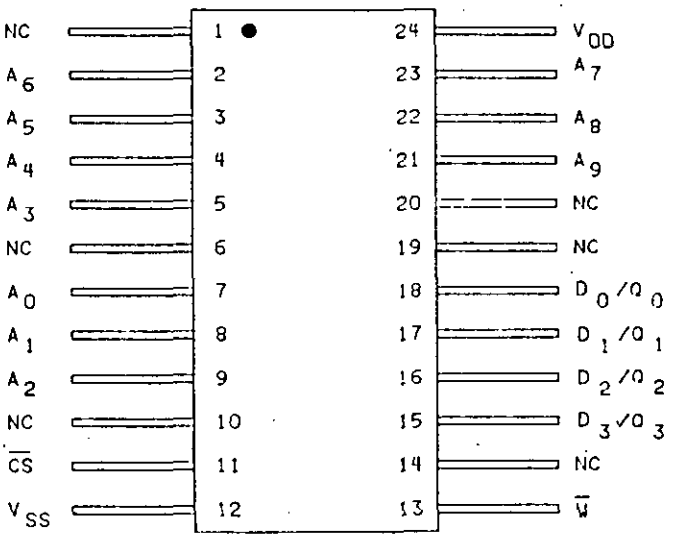
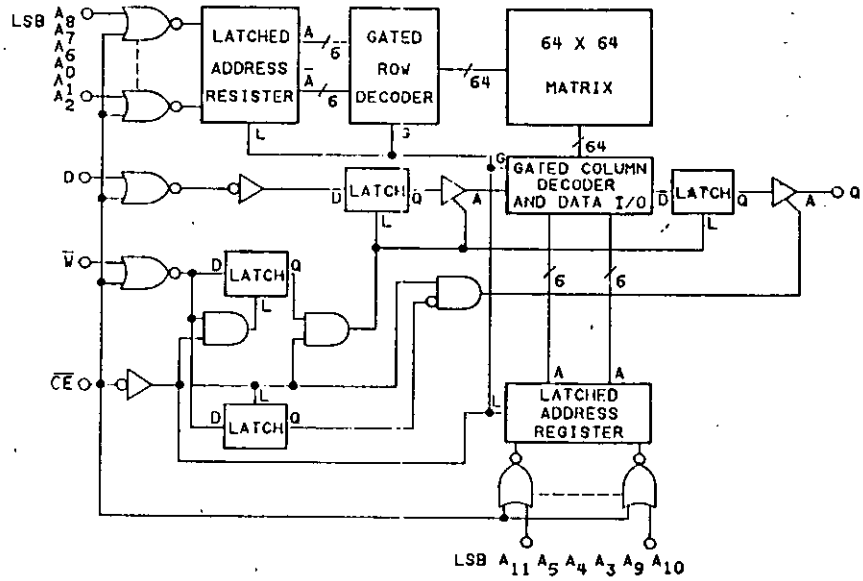


FIGURE 1. Terminal connections.

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Device types 01 and 03



Device types 02 and 04

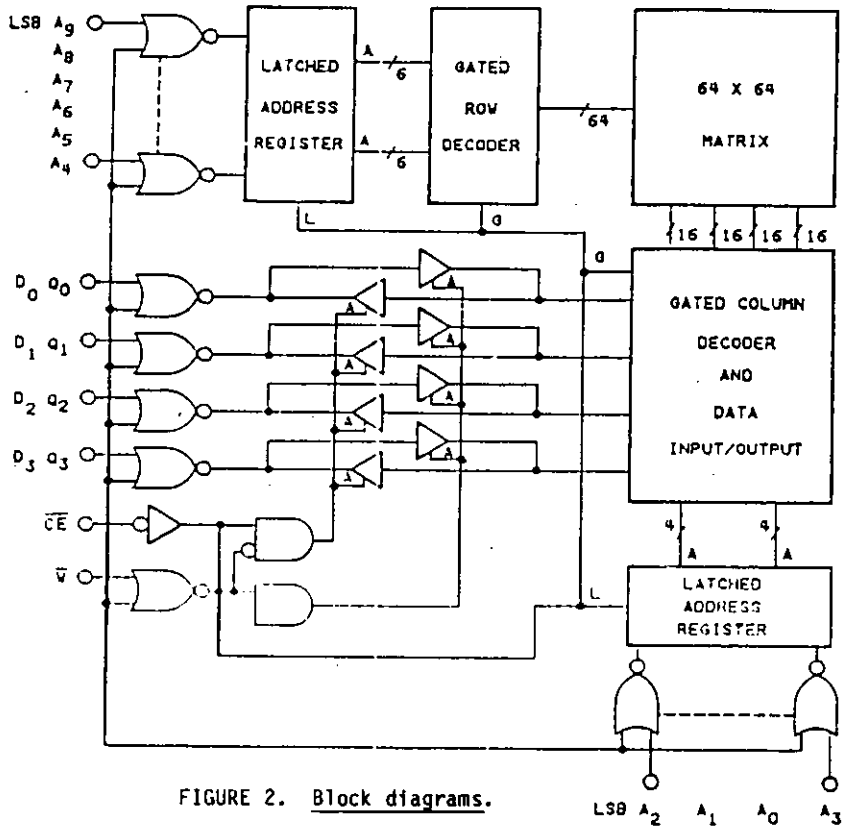
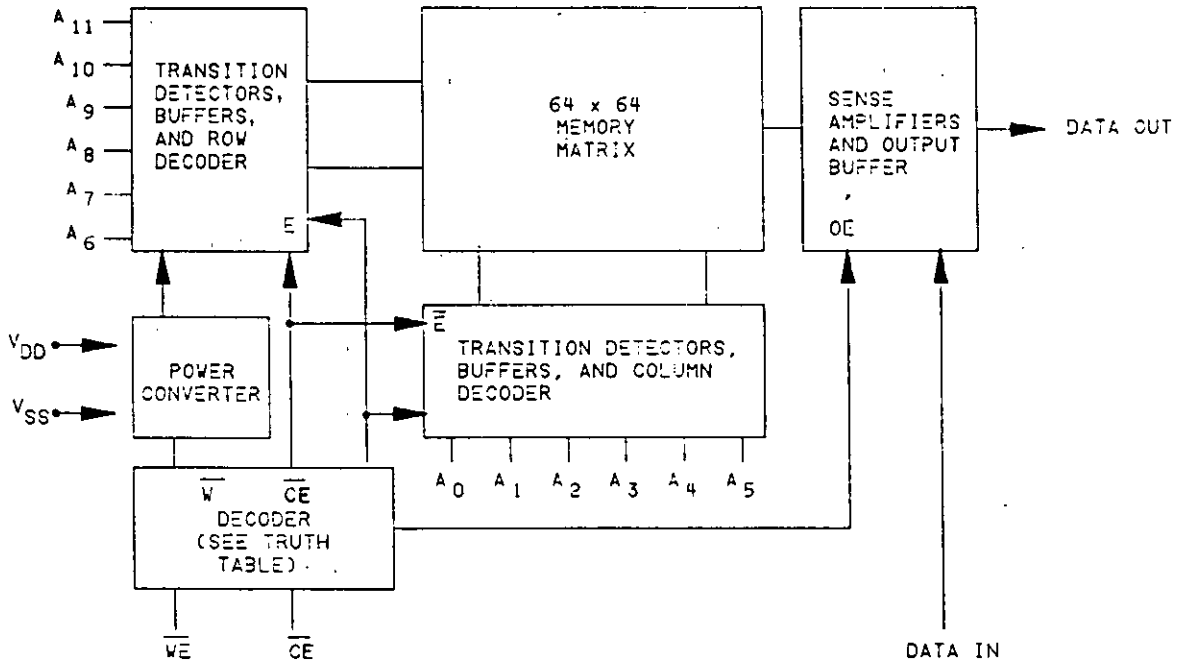


FIGURE 2. Block diagrams.

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Device type 05



Device type 06

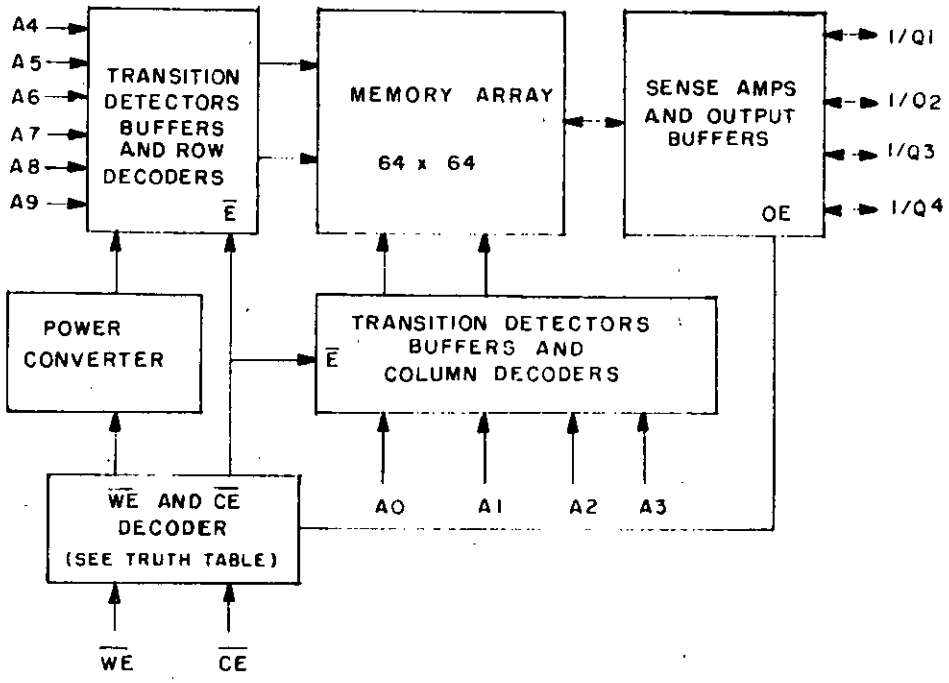
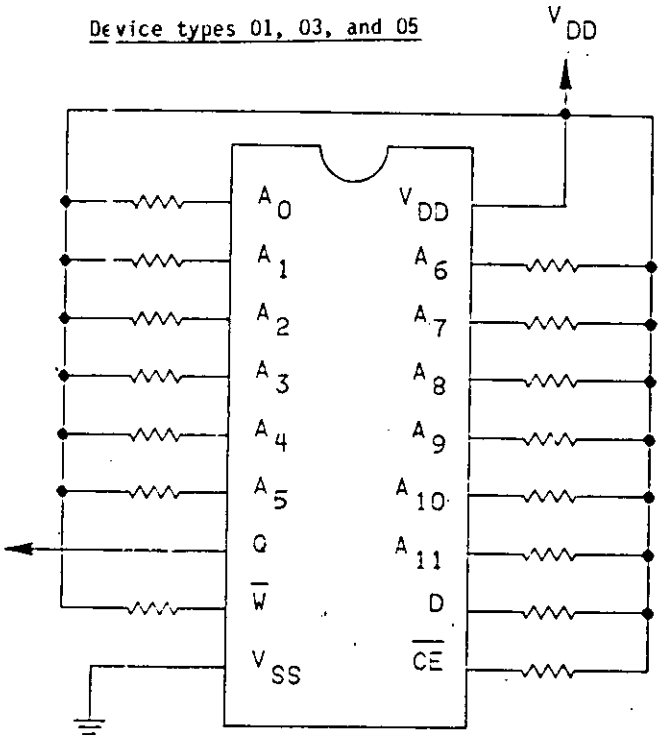


FIGURE 2. Block diagrams - Continued.

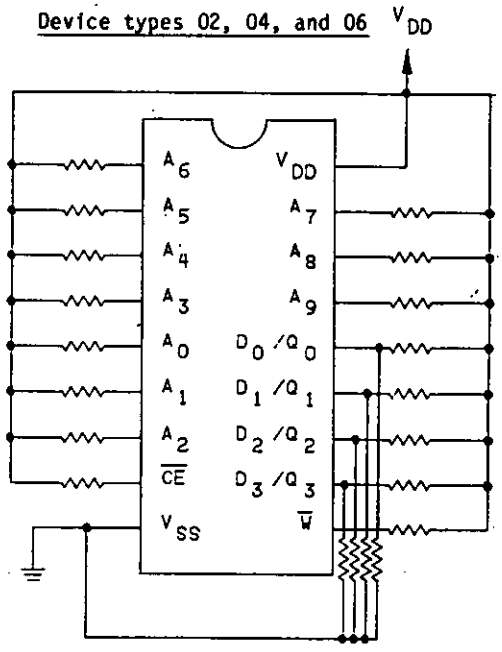
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Device types 01, 03, and 05



Device type	Q min
01	$\frac{V_{DD}}{2}$
03	$\frac{V_{DD}}{2 \bullet 5}$
05	$\frac{V_{DD}}{2}$

Device types 02, 04, and 06



Device type	V_{DD} min
01	5.0 V
02	5.0 V
03	6.0 V
04	6.0 V
05	6.5 V
06	6.5 V

NOTES:
1. All imputs are connected through a 1 kΩ to 47 kΩ resistor.
2. 05, 06: For static II burn-in change input connections to V_{SS}.
3. 05, 06: All 1's pattern filling array for static I.
4. 05, 06: All 0's pattern filling array for static II.
5. For device type 06: The pull down resistors on the outputs are not required.
6. Output pin device types 01, 03, and 05: Connected either through a resistor to a power supply or directly to a voltage divider accross the supply rationed as shown.

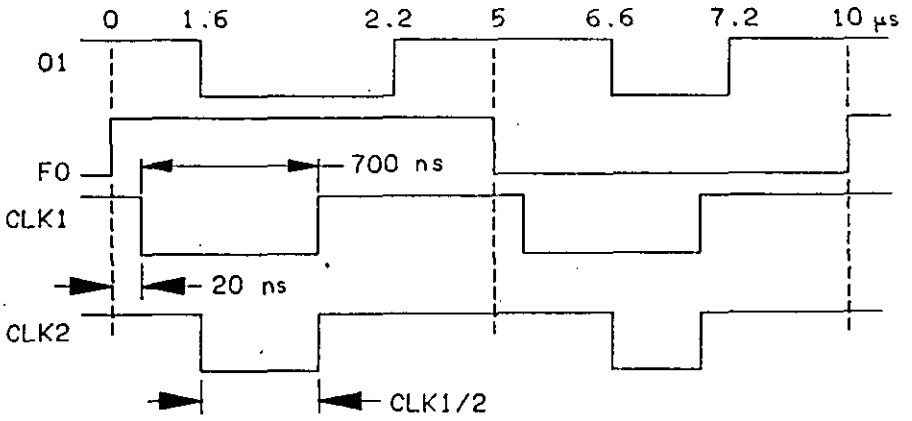
FIGURE 3. Static burn-in circuit.

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Terminal connection chart (see notes 1 and 2)

Device type	V_{DD} min	A_0	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	A_9	A_{10}	A_{11}	$\overline{CE}$	W	DO/Q_0	D1/Q_1	D2/Q_2	D3/Q_3	Q
01	5.0 V	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13	F14	F0	F1	F2				F2
02	5.0 V	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12			F0	F1	F2	F2	F2	F2	
03	6.0 V	F0	F1	F2	F3	F4	F5	F6	F7	F8	F10	F11	F12	CLK1	CLK2					F9
04	6.0 V	F4	F5	F9	F3	F2	F1	F0	F6	F7	F8			CLK2	CLK1	F10	F11	F12	F13	
05	See note 3	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F13	01	F12				$\frac{V_{DD}}{2}$
06	See note 3	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9			F14	01	F10	F11	F12	F13	

NOTES:
1. F0 = 100 kHz minimum, F1 = 1/2 F0, F2 = 1/4 F0 ect.



2. Resistors: (all values ±5%)
   Type 01, 02: All input and output pins 47 kΩ.
   Type 03, 04: Same burn-in board both type, with pins 7, 11, 12, 13, and 14 connected through 27 kΩ resistors to frequencies shown in table; 300Ω resistors on other pins.
   Type 05, 06: Pins connected through 2 kΩ to 15 kΩ, depending on pin function, as indicated in table.
3. V_{DD} min = 6.5 V for dynamic burn-in; V_{DD} min = 5.5 V for steady-state life test.

FIGURE 4. Dynamic burn-in and steady-state life test circuit.

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for devices intended to be marked as radiation hardened. When group E testing is performed, it shall be in accordance with table V of method 5005 of MIL-STD-883 and 4.5.5 herein.

4.4.6 Inspection of packaging. The inspection of packaging shall be as specified in MIL-M-38510.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

NOTE: In order to prevent "latch-up" during device turn-on and also prevent a high initial operating current (I_{DDOP}), set all device terminals equal to V_{SS}. Raise V_{DD} and $\overline{CE}$ simultaneously to normal system operating V_{DD}. Do not allow V_{IH} at $\overline{CE}$ to exceed V_{DD} +0.3 volts while rising. Not necessary for SOS products.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit V_{SS} terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 Life test, burn-in, cooldown, and electrical test procedure. When devices are measured at +25°C following application of the steady state life or burn-in test condition, all devices shall be cooled to +35°C prior to removal of bias voltages. Any electrical tests required shall first be performed at -55°C or +25°C prior to any +125°C tests that are required.

4.5.3 Delta measurements. Delta measurements, as specified in table II, shall be made and recorded before and after the required burn-in screens and steady state life tests to determine delta compliance. The electrical parameters to be measured, with associated delta limits are listed in table IV.

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TABLE IV. Delta limits at +25°C.

Parameters 1/	Limits 2/			
	01, 02	03	04	05, 06
I_{DD}	+5 µA	+15 µA	+25 µA	+30 µA
V_{OL}	+60 mV	+60 mV	+60 mV	
V_{OH}	-200 mV	-400 mV	-400 mV	
I_{OL}, I_{OH}				±10%
I_{OHZ}, I_{OLZ}				±500 nA
I_{IL}, I_{IH}	±50 nA	±100 nA	±100 nA	

1/ Conditions and mechanization of measurements shall be as specified in table III, subgroup 1.
2/ Delta limits apply to an increase from the initial value (e.g., pre-life test I_{IH} = -120 nA, post-life test I_{IH} = -170 nA).

4.5.4 ESDS testing in accordance with MIL-STD-883, method 3015, will be performed as part of group B at initial qualification and after any processor design changes affecting input protection circuits.

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